Health Policy and Technology (2017) 6, 292–301

Available online at www.sciencedirect.com

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journal homepage: www.elsevier.com/locate/hlpt

Dynamic capabilities in e-health innovation: Implications for policies

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Available online 24 February 2017

KEYWORDS
E-health;
Municipalities;
Policies;
Dynamic capabilities view;
Process orientation;
Business process management

Abstract

Objective: To mitigate the effect of caregiver shortage, collaborative networks in Norwegian municipalities are exploring the possibilities provided by e-health and welfare technologies. However, extracting benefits from such technologies depends on many factors.
Methods: In this study, an extensive literature review is performed to compare e-health and other sectors in terms of the critical success factors in collaborative business process management. Using the dynamic capabilities view as a general theoretical lens, and a process orientation framework for operationalization, these factors are then conceptualized and validated in a cross-sectional study of cases in the Norwegian municipal e-health sector.
Results: The study contributes to e-health research by identifying the key factors that influence performance. These factors are significantly driven by government policies and regulations. Our findings challenge the assumption that welfare technology networks can be built from the bottom up without government intervention. Regulatory interventions are needed, to obtain process performance metrics and foster viable, long-term business models for the participating institutions.
Conclusion: The findings have an impact on research and practice, especially in local public management, for predicting and prescribing future development in this context. There are indications of significant gaps in government policies and regulations. Further research should examine whether and how these findings transcend the chosen context.

Rationale

In this section, we present the motivation behind this paper, the problem statement, address the gaps in the literature, as well as important challenges for practitioners, and introduce the expected contributions from this study.

Motivation

As the aging population is increasing in many industrialized countries, there is a need for new technologies in health-care (e-health) and improvements in management in the form of better coordination of processes. This study is part

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http://dx.doi.org/10.1016/j.hlpt.2017.02.003
2211-8837/

of a larger research project about the role for collaborative business process management (BPM) in e-health innovation, aimed at meeting these challenges. We can look at business process management (BPM) as a management strategy [1], supported by integrated tools, ranging from business process modeling notation as an illustration of as-is and to-be scenarios to BPM systems with full workflow integration using service oriented architecture, integrated in an enterprise architecture [2]. Enterprise architecture can be instrumental in implementing the business model (value creation, delivery, and capture) [3], in advanced enterprises. Despite the growing interest in BPM, there is a gap in the literature regarding the importance of context for successful BPM [4], including the inter-organizational context in which BPM is coordinated between two or more separate legal entities. As a result, uncertainty exists about how general BPM research can inform collaborative efforts in highly specialized contexts, such as the health sector, where demographic changes pose new challenges in elder care and forces collaboration in the production of healthcare services in most industrialized countries.

Problem statement

Aging populations in many industrialized countries, with increased numbers of people in need of care and a diminishing ratio of available caregivers, call for efficient health care supported by e-health technologies.

In Norway, the responsibility for primary care is placed on municipalities, through the Coordination Reform [Samhandlingsreformen] [5]. This division of roles was formalized as a legal act in 2011–2012. Hospitals are state-owned and managed through large regional health enterprises. Many municipalities are small, with the average population size in 2015 at around 11,500. Municipalities often collaborate to leverage joint resources in e-health pilots, establishing new work processes for caregiving at home or in institutions. These networks, such as the Southern Junction [Knutepunkt Sørlandet] [6], span collaboration in several areas, for example, a common climate plan, joint area planning and transport cooperation, local medical services, personnel restructuring plan, IT coordination, and future cities. In business management science, this can be termed horizontal collaboration. Such collaborative networks are also leveraged in negotiations with regional hospitals and the national government. This type of collaboration is called vertical or upstream collaboration. Horizontal and vertical collaboration involve aligning work and (digital) information processes across organizational borders.

Local e-health innovation in Norway largely involves home care and health monitoring products (welfare technology networks). These development projects are encouraged by the national government. However, according to government-funded reports, this development seems to build on the assumption that it can be undertaken locally in a bottom-up approach without intervention from the national government [7].

Contribution

Research results in Norway and abroad show that such innovation initiatives often partly fail; the benefits are sometimes not what was expected, progress is slow, and the projects run into chasms as the initial funding ceases and the projects fail to go into sustainable production. The performance antecedents, drivers, and barriers in such projects represent an emergent theme in (management) information systems research. There is a gap in the literature regarding the key success factors for performance and strategy alignment in e-health at the inter-organizational level of analysis and the mutual influence between these factors. At the same time, practice in this context seems to ignore the process management literature. A recurrent theme in this literature is the need for senior management involvement [8,9] as a prerequisite for success with the business process redesigns that these e-health projects often entail. In a public setting, such as the present study context, this means management involvement from the local and national government. Through this study, we aim to question the governing practitioners' assumption [10,11] that welfare technology networks can be successfully built locally from the bottom up.

The research questions (RQs) are as follows: (1) What key factors (drivers and barriers) influence e-health service performance in general and in the chosen context (service, meaning the consumable outcome of the processes)? (2) How do they influence one another? (3) How do the municipalities perceive the e-health performance, and what are the implications for policy decision-making? Regarding RQ1, the main proposition for this study is that process orientation is a key factor, and at the same time, an antecedent to the application of integrated BPM methods and systems [12,13].

To answer these questions, we applied the following methods, stepwise: a literature review, concept operationalization, and interviews. The literature review was an extensive literature study on inter-organizational collaborative business process management [1] in e-health and in other comparable sectors. The articles deemed relevant according to the inclusion criteria were analyzed and categorized. This step yielded 36 critical success factors (CSFs), which, in turn, were collected into eight concepts. Related works by one of the article's authors [14] provided the ninth concept. We then operationalized the concepts we found as a guide for the semi-structured interviews (see the appendix) using the dynamic capabilities view [15] and the process orientation frameworks [16] as the theoretical lenses. We interviewed employees from three municipal e-health collaboration networks, using the guide from step 2. Based on this, we developed and proposed a consolidated model with relations between the CSF concepts. By looking at three cases in step 3, we can compare and look for generic concepts and eventual contrasts, thus underpinning our final model. We found nine performance-impacting concepts in step 2 and further explored and validated (as existing and significant) them using interviews in step 3.

This study contributes to the literature by enhancing the knowledge on e-health collaboration and revealing critical factors that influence performance, as well as insight into the relations between these factors. This study also relates the domain capabilities view (DCV) to levels of analysis in a novel way.

The rest of the paper is organized as follows: We explain the methods and process of the data collection and analysis.

Then, we discuss the findings, focusing on the similarities and differences between the cases. We conclude by addressing the research questions with the findings and discussing the implications for further research and practice.

Methods

In this section, we describe in more detail the methods we used for conducting a review of the literature and how findings were followed up by interviews in context to confirm or elaborate on the findings from the literature. We also present in more detail the specific cases of collaboration we studied in the context.

Literature review

We organized the literature review in three steps. The first step involved 10 searches in several databases (Scopus, IEEE, Emerald, and ISI Web of Knowledge), with the following key search terms in the title, keywords, or abstract: "collaborative business process management", "BPM" and "inter-organizational". We used different truncations of these terms and in different combinations, such as ("business process"), ("management"), ("collabo*" OR "coop*"), or ("inter-org*"). We also conducted specific searches focused on BPM in information systems research (IS) sources, such as *Business Process Management Journal* and *Journal of Enterprise Information Management*. We also performed a specific search on "BPM" in the proceedings of the Hawaii International Conference on System Sciences, as this conference has had a large track of BPM-related research.

We included only articles in English that were published after 2004. The last search was performed on 30 June 2013 and returned 5361 titles. A screening of the articles showed that the majority were irrelevant to the research questions. We excluded articles that did not describe inter-organizational process collaboration or success perspectives or factors, such as articles that focused solely on technical feasibility. This screening reduced the number of relevant articles to 47. After we inspected the results from the two previous steps, we considered some articles were particularly relevant. We used these articles in the last step in the search process, in which we combined forward and backward citation searches for articles that cited or were cited in these articles, as recommended by Webster and Watson [17]. In this step, we included articles published before 2005. We identified three additional interesting articles, for a total of 50 articles (see the appendix). Eleven of the 50 articles reported on studies in the e-health context (see the appendix: C8, C9, C16, J1, J3, J5, J7, J11, J12, J15, and J28).

Concept operationalization

The literature review, accompanied by earlier research into the context, a comparison of e-health research and innovation in Switzerland and Norway [14], presented a challenge: The factors related to alliance orientation (AO) and process orientation (PO) seemed to live separate lives in two levels of the analysis, based on the reported narratives of the cases in which they were elicited. The AO critical factors seemed dependent on the will of senior managers in the domain and the challenges of strategy, politics, and culture on an inter-organizational or even societal level of analysis. PO (intra-organizational) seemed to reside more at the lower-middle management or project management level, in the less strategic and more operational and tactical domain between organization and IT systems at an intra-organizational level of analysis. How were the two related? How did they influence one another? To clarify this, we chose a cross-sectional interview study to validate the existence of the two concepts (AO, PO) and explore the relation between these two and other concepts.

We gained access to an interesting mix of cases: networks consisting of large (Easttowns), small (Rivervalley), and medium (Fjordbanks) municipalities with different local government administrative resources available, according to size, thus highlighting the concept of the structure or the meaning of relative organizational size. The networks were measured according to the population size on the Norwegian scale. Within these networks, we focused on the e-health projects that we, and the interviewees, deemed the most significant.

Analyzing the findings from the literature review, we identified Teece et al.'s DCV [15] was relevant for explaining alliance orientation factors and the overall view. Teece et al. claimed that performance and competitive edge come from their (general) "processes", shaped over time by the enterprises "paths", and "positions" (or "positional assets"). Paths in this model consist of "path dependencies" and "technological opportunities". Positional assets consist of the coherence of internal and external processes and incentives (encompassing process orientation [16], BPM the as method or system, etc.) and specific assets, such as technological assets, complementary assets (commercialization), financial assets (dependent on the firm's balance sheet), reputational assets, institutional assets (e.g., regulatory frames), market structure assets (e.g., market position), and organizational boundaries (internal integration vs. contractual arrangements).

Processes, meaning higher-order processes to distinguish them from the operational business processes in Kohlbacher and Gruenwald [16], consist of coordination or integration; (could be the realized/operationalized PO/BPM); learning (a dynamic concept) – "even more important: The concept of dynamic capabilities as a coordinative management process opens the door to the potential for inter-organizational learning. Researchers (...) have pointed out that collaborations and partnerships can be a vehicle for new organizational learning, helping firms to recognize dysfunctional routines, and preventing strategic blindspots" [15]; reconfiguration; and the ability to rapidly adapt the enterprises resources to meet new situations. Thus, the DCV model offers an explanation and prediction for how enterprises mobilize resources through collaboration and gradually transform them into BPM capabilities through learning.

The DCV model has had much impact as measured in the number of citations and has been expanded through Sambamurthy et al. [18] and reconfigured as new models in Wu et al. [19]. Wu et al. introduced the concept pressure, which could be an analogy for the Coordination Reform [5].

We used Kohlbacher and Gruenwald's conceptual framework [16] to explain and measure process orientation in this

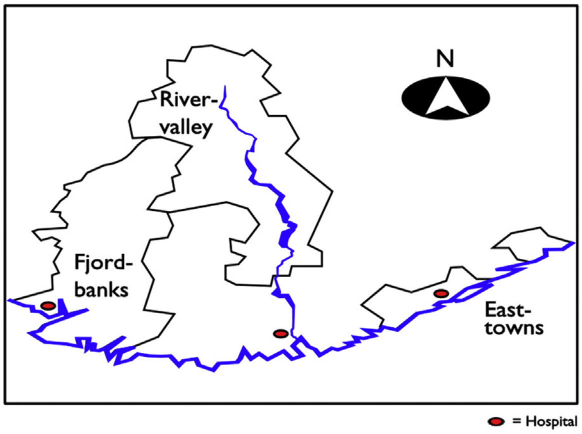

Fig. 1 Outline map of the three collaborative networks in the study.

context. Process orientation details the positional assets in the dynamic capabilities view [12,15]. We loosely followed the framework when we designed relevant parts of the interview guide.

Interviews

To study the relations between the main factors identified in the literature review, we decided to perform an interview study in the context of Norwegian collaborative municipal e-health innovation. This decision was based on the fact that in Norway, municipalities bear the overall responsibility for primary care, ref. the problem statement of this article. We performed the cross-sectional interview study in the fall of 2014. We conducted three semi-structured interviews with e-health middle managers and project managers who represented three municipal networks in Norway and participated in various ongoing e-health implementation projects (see Fig. 1).

We constructed an interview guide (see the appendix) to cover all the relevant factors in this study, using the theories and framework discussed. We loosely followed the guide to allow for follow-up questions.

We recorded and transcribed the interviews. We then sent the transcriptions back to the interviewees for additional comments and corrections. The interviewer translated the approved narratives from Norwegian to English and used pseudonyms for confidentiality and a critical examination. The narratives were imported into NVivo10 for coding. The coding was done top-down (using the pre-set concepts from the interview guide to form nodes) and bottom-up (eliciting new nodes from the material). We examined the coding and the grouping of nodes independently and then discussed the group structure and aggregation of the nodes until we reached agreement.

Case contexts

The contexts and projects are as follows (real names have been replaced with pseudonyms): Easttowns, Fjordbanks, and Rivervalley. In Easttowns, the Digital Night Supervision project is a joint project between three urban municipalities that are part of a larger network, Easterncounty, with a total of six municipalities. Fjordbanks is a network with six municipalities (joint projects: Better to Own Life and Security Package). Rivervalley is a network with four municipalities (joint project: reducing the number of alarm centers). The local medical services (LMS) office is working on projects such as coordinating the home and personal safety alarm centers into one inter-municipal center. The LMS office also works with e-health interoperability issues. The cases are very different in area size and population density. Thus, they address the question of how the organizational structure (size and resources) affects the distribution of services.

Results

The interviews validated the existence of the concepts found in the literature review and previous studies. There were some differences between the informants in the three cases in attitudes toward the affordance of welfare technologies but also similarities, such as in the perceptions of the impact of government interventions.

Literature review and conceptualization

From the literature review, we found 36 factors that we grouped into eight main concepts (see the appendix). We also found that some factors were transformed during phases or stages in the life cycle of a project. For instance, successful inter-organizational management had to be open and inclusive in the early implementation stages and then had to be more disciplined and coordinated in the later rollout and scaling up stages. The main concepts are as follows: alliance orientation, process orientation, business model orientation, government policy orientation, structure (organization size), maturity, which can be measured with maturity models, such as the capacity maturity model (CMM) [20], technological diffusion of the innovation, and digital information infrastructure. A short explanation of these eight concepts, and citations, is shown in the appendix.

A comparative study of state-of-the-art e-health innovation in two European regions, conducted by Garmann-Johnsen [14], indicated a ninth factor: problem (and room for solution) perception. This concept entails that choice of themes for the local e-health research and development pilots in both countries are clearly influenced by varying structures in the general organization of healthcare strategies, the distribution of hospitals, as well as by national government-sponsored programs and their focuses. The comparative study [14] added a new factor to the concept of structure: It is dependent not only on the size of the focus organization or network but also on the focus organization's relative organizational distribution and size vs. upstream partners such as regional hospitals.

Findings from the interviews

The cross-sectional case study confirmed the existence of the factors found in the literature and related research and shed new light on the relations between these factors. The factors that we found in the interviews that influence performance are detailed. The local areas covered were Easttowns (the two areas to the right, in Fig. 1; three municipalities that cover a total of 767 km^2 and have 68,000 inhabitants; inhabitant numbers are given here as reported in the interviews), Fjordbanks (six municipalities, 4175 km^2, 36,000 inhabitants), and Rivervalley (four municipalities, 4495 km^2, 7500 inhabitants).

For the Easttowns project, Izabella Sagan (Kensbay) and Arnie Whiteman (Jasonsvalley) were interviewed on 22 October 2014 at Jasonsvalley Town Hall. Izabella and Arnie are two of the three local project managers for the Digital Night Supervision e-health project. Arnie heads the project manager group. The three municipalities have a total population of about 68,000. They are part of Easterncounty, a larger regional collaboration network of seven municipalities. For the Fjordbanks region collaboration, Marsha West and Gwen Owings were interviewed on 29 October 2014 at Rivermerge Municipality City Hall. Marsha works in the Rivermerge Unit for Livelihoods, the case management office for health and social services. Gwen is the project manager for Better to Own Life (BtOL) in the six Fjordbanks municipalities. The Fjordbanks collaboration consists of six municipalities with a population of about 36,000. For the Rivervalley region collaboration, Tom Jackobson was interviewed on 24 October 2014 at Raftingwater (the host municipality for LMS, a joint office for four municipalities). Tom is responsible for information communications and technology (ICT; health care systems). The Rivervalley collaboration consists of four municipalities with a population of 7500.

First, we wanted to identify the subject for the e-health collaboration between the municipalities. Examples from the transcript (Q = Questions, A = Answers):

> Q: What is the biggest welfare technology project you are working on now?
> A (Easttowns): Night Supervision with technology support … Running problems have been related to logistics and general inertia in the system. This is a new task that comes on top of everything else for those responsible for supporting systems.
> A (Fjordbanks): Better to Own Life is a welfare technology project under the Health Network Fjordbanks collaboration. Residents who receive such help could prolong their life expectancy and stay safely at home longer, based on the challenges we know are coming. We must change our way of working and way of thinking. This project works with attitudes and motivation … We also have a security package. Connect (Nordic Project) includes best practices in the implementation of welfare technology projects, and the EU project features the world's best care homes. These interlock.
> A (Rivervalley): Safety alarms. Here, the number of alarm centers is reduced. The system must also be replaced over time [Telenor phase out of analog networks, probably caused by economic considerations] by mobile-based or broadband systems. (But it is not always easy to install broadband in every home. In some homes, it is not possible.)

Alliance orientation
All three cases reported a high degree of alliance management capabilities within their respective municipal networks. They all have the appropriate structure for facilitating collaboration on several levels, with fixed meeting points and forums. Senior management appreciates the collaboration efforts; overall, the networks and joint projects receive high attention from all participating municipalities.

Process orientation
Process performance is a strongly emergent performance-driving factor, but the state-of-the-art e-health innovation has gaps in goal orientation. We found clear signs of all the factors associated with process orientation in the case studies: a growing use of structured systems, systematic user involvement, coherent project management methods, rigid procurement procedures, clear process ownership, a high degree of management involvement, a certain flexibility of processes, high co-worker involvement, and a clear understanding of the present (as-is) situation. For instance, Fjordbanks was in the process of systematic process modeling for the current workflows. What seemed to be missing was a clear goal. With the exception of simple numbers for plant implementations of different products, key performance indicators or measureable benefits in terms of savings or increased productivity seemed to be missing. In another case (Rivervalley), the interviewee did not perceive any vision for health care. The apparent lack of goal orientation may be consistent with the weak but growing awareness of business models.

Awareness of business models
The municipalities' e-health pilots will face a challenge when the current national co-financing ceases (as proposed in the national government budget for 2015). There is a growing awareness of the need for viable business models. There seems to be a gap in the inter-organizational top-down approach to business change, as the municipalities enter a complex market of different finance and reimbursement models. Another issue is that municipalities as organizations are not used to thinking in terms of markets and business models. For instance, when an Easttowns interviewee was asked what would happen when the project funding stops, he/she replied that the handover to daily operations remained unclear.

Policies: the role of the national government
A surprising discovery from the interviews was the significance of the national government in correlation with large gaps in regulations and national politics. On the positive impact side, the government promotes alliances, teaches good project management practices, and funds e-health pilot projects through national programs. On the negative impact side, the national government governance of the local health care system is unclear: Regulations do not provide municipalities with finance opportunities through deductibles or government reimbursements. Central information infrastructures are not yet ready. These gaps can be serious barriers to success and performance in local e-health projects. National government regulations and politics have an impact on all the factors that influence performance; thus, they have a significant indirect impact on performance.

Structure: size of the collaboration networks

The interviews provided some support for the assumption that "bigger is better" when it comes to the size and performance of municipal organizations. For example, municipalities that have a critical mass seem to be able to mitigate the challenges posed by gaps in the information infrastructure (as shown in the Rivervalley case). Larger municipalities or municipal networks also seem to have more resources for project work, driving innovation. This can be termed ambidexterity, or the ability to maintain control of operations and development.

The question of municipal structure is a question of political significance in Norway, as the number of municipalities will be reduced as a result of an ongoing reform. However, tradition and attitude toward innovation also play a role in e-health performance, according to the interviewees. Some of the interviewees (Fjordbanks) questioned the need for bigger municipalities and pointed to efficient collaborative networks as an alternative.

Organizational learning: the mechanism of process maturity

When the interviewees were asked about the most important lesson learned from the projects or what advice they would give others attempting to follow the same road, the common answer of all three groups was the significance of embedding achieved knowledge in the whole organization. Consequently, much effort is exerted in disseminating the project results in their own networks and in open forums. The importance of these learning cycles is directly related to overall performance, inter-organizational collaborations, and process orientation. In this respect, the interviews clearly confirmed this study's assumption, based on Teece et al. [15].

The following quotes illustrate this point. An Easttowns interviewee stated, "The problem is the allocation of responsibility—who feels responsible for new tasks? Often, the project manager must follow up [issues that arise]. You learn as you go." A Fjordbanks interviewee stated:

> We have two show homes: Heathvalley [another municipality] and Rivermerge. Here we bring users to view [different smart home and welfare technologies]. It is something else to see it!
> Who does what with operational measures and alerts? Allocating responsibility requires collaboration across departments and units.
> Now every municipality [in Fjordbanks] is in the process of creating a process map from A to Z, from the needs survey through acquisition to service and maintenance, etc. This will clarify roles and responsibilities.

E-health diffusion of innovation

The interviews painted a picture of technology adoption almost exclusively at a very early stage. Thus, there is little longitudinal empirical data to report. One informant from Rivervalley, citing the National Audit Office [Riksrevisjonen] [21], doubted the inherent value and benefits of some of the welfare home technologies being piloted with funding from national government programs.

Digital information infrastructure

Together with the large impact of government policies, all three cases emphasize the significance of an efficient digital information infrastructure system (DIIS) and standards in e-health. This also has a close connection to the roles of national government bodies. One informant from Rivervalley envisioned the potential for DIIS to underpin business growth by allowing a connection to new third-party systems and products. This again is an area where an active government role is needed. According to the Rivervalley informant, "The information infrastructure is the key. Smart home [technologies are an] affectation."

Problem and solution perception

Garmann-Johnsen [14] proposed that the general structure and division of labor within health care, such as the hospital structure, has had a large impact on problem perception within e-health in Norway. This study also showed that the national government plays a direct role in the municipalities' e-health affairs and that there are gaps in the form of unanswered questions ranging from ethics to laws and regulations. This issue became even clearer in the interviewees' replies regarding the role of the government: Q: Are there dilemmas of ethics/injustice [when it comes to who receives and does not receive new services]? A (Fjordbanks): "You must create guidelines. Should we start with the Health Care Act? This is not quite in place. The criteria now are for qualifying for home care."

Regarding budget frames, the Fjordbanks interviewee explained that the county governor [Fylkesmannen] will take over the role that the Welfare Program has had in financing municipal e-health projects:

> If the county governor and discretionary funds [skjønnsmidler] take over the national welfare technology program [to be abolished according to the state budget proposal as of 14 October], they then get a double role: supervision and innovation. What should then be the criteria/keys?
> Q: I guess we have some time [to sort out the real questions and appropriate answers] before the elder boom arrives.
> A (Rivervalley): Yes, but we must also think about a public undertaking. When can I cancel my contract with Verisure [real name of a private alarm company] and get a public alarm? Municipalities have financial constraints.

Perception of performance

The interviewees showed great confidence and pride in the intermediate results from the ongoing pilots. The products are technically feasible; they work as they are supposed to. Municipalities that have experimented with home care and monitoring technologies have collected much knowledge on the benefits perceived by the end-users or patients. Another significant impact of the early results and experiences with implementing new e-health technologies is learning: a growing awareness and knowledge about changes in work processes, routines, and procedures.

Examples of excerpts on goal perception and orientation illustrate this point:

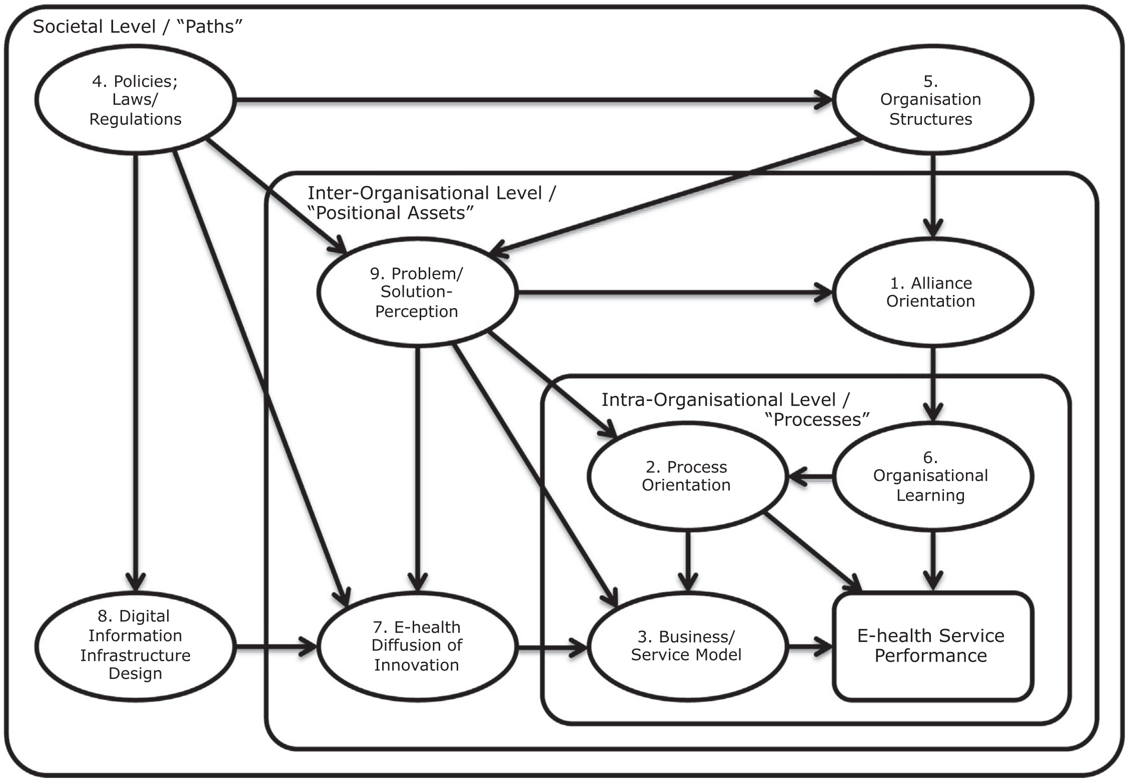

Fig. 2 Proposed multi-factor model of performance in the Norwegian municipal e-health context based on the dynamic capabilities view. See the appendix for details of the factors in the literature.

> Q: Do you have any goals for the project, or are they not specified? A (Easttowns): The project application was in response to a national call for proposals, which stressed that the welfare technology should be an integral part of care by the year 2020. The goal may be to curb spending growth in the longer term ... In the short term, it is not possible to rationalize more. A night watchman keeps, for instance, a list with 40 apartments; you must have at least one watcher. The main objective therefore is improved quality and safety.
> Q: What have you accomplished so far? A: We have achieved what we wanted: We have installed equipment, and it has worked as expected.

Examples of excerpts on the perception of benefits or economy or savings as a result of introducing new technologies:

> Q: Are there fewer visits in homes? A (Fjordbanks): In Rivermerge, they cut out a night watchman in home-based services [housing for handicapped]. It has provided better services—it was almost monitoring what you were doing. Users would be alone sometimes. Marsha: We want "security without interference."
> Gwen (on other benefits and patient experiences): A user has FaceTime [mobile video conferencing], summons staff, but can otherwise be quite alone—something he could not be before. Another user got a door sensor. He will not have to move or sleep in a common area. So, it was a win-win for the municipality and the users.
> Q: [What are the benefits of] collecting receiving alerts? A (Rivervalley): Easier maintenance of ICT boxes—one box instead of several. Otherwise, there are no work-related changes.

Discussion and conclusions

In this section, we review the findings in light of the research questions. We then discuss the study's contribution to theory and practice and suggest areas for further research. The e-health pilots induce bottom-up learning in participating organizations. This explains the transition from alliance orientation to process orientation. This explanation is also in compliance with the DCV model [15].

This learning has to be met by top-down governance to provide resilient results. Most studies on inter-organizational collaborative process management see such collaboration as largely free from competition and conflict (cf. the appendix, citations; C2, C17, J2, J9, and J12). Furthermore, because health care is a public domain, there seems to be an assumption of political alignment and no coordination problems. However, a recent case study in the Norwegian municipal e-health context has shown that this is not the case [22].

Provan and Kenis [23] documented the importance of governance models for networks that interact with public actors as in e-health and claim that under certain conditions, such as low confidence, different goals, increased need for expertise, a centralized management, and governing are the most effective. This is in compliance with the DCV [15] and its predecessor the resource-based view theory [24], in which similar circumstances call for internal integration as a more efficient mode for governing than contractual arrangements.

Key factors influencing performance

The interview study strongly indicates that all the key concepts of factors found in the literature review and the comparative study [14] (see Fig. 2 and the appendix for more details) play a role in municipal inter-organizational collaboration in e-health.

Our general impression from the cross-sectional interviews in the three e-health cases is that the similarities between the three cases can form the basis for some general propositions. Differences may come down to contextual and almost personal attitudes, such as toward the affordance of welfare technologies in general. Where the three cases show great similarities in their answers, such as in the view of the importance of government interventions, the findings are generalizable at least within the Norwegian e-health context. In particular, the strong indirect and influential role of national government policies, as well as uncertainties surrounding the future funding of welfare technologies, raises the question of whether a political gap needs to be filled.

More specifically, we propose that the national government must impose additional financial incentives. If we draw on Garmann-Johnsen's comparison with Switzerland, which has a financial system in place [14], this seems to underpin this need as a national, and possibly international, generalizable finding of positional assets as a critical (and in the Norwegian context, missing) success factor for e-health innovation. However, further research is clearly needed here.

Relations between the key factors

The DCV goal of achieving long-term competitive advantages through less costly processes [15] refers to the familiar challenges of health care for an aging population, lack of caregivers, and rising costs. The goal also includes a process perspective and understanding of the process as an additional capability that has proven suitable for coordinating processes and services across operators [12].

These perspectives are summarized in a testable model that we propose based on the perspective of the DCV, the literature review, and the interviews (Fig. 2). The concepts that influence performance mainly residing at the societal level of analysis policies (Fig. 2; path 4), structures (Fig. 2; path 5), and digital information infrastructure (Fig. 2; path 8) can be cast as new paths in the DCV (15). The inter-organizational level introduces new positional assets, alliance orientation (Fig. 2; path 1), e-health diffusion of Innovation (Fig. 2; path 7), and problem/solution perception (Fig. 2; path 9). Finally, all this shapes the organizations' processes: process orientation (Fig. 2; path 2), business/service model (Fig. 2; path 3), and organizational learning (Fig. 2; path 6). Alliance orientation (Fig. 2; 1) matures the organization toward higher levels of process orientation (Fig. 2; path 2) through organizational learning (Fig. 2; path 6).

Municipalities' perception of e-health performance and policy implications

Perceptions are detail-focused. The case studies point to fragmented benefits, but a holistic goal orientation for the municipalities' future role in primary care seems to be missing or has not been clearly communicated to the middle managers interviewed in this study. The municipalities' perception of e-health performance seems to be based on assumptions of benefits that are not quantified. It may be heavily influenced by government reports and policies. Government reports [25] seem to assume that welfare technology has an inherent value, without giving any directions for how to achieve and manage the assumed benefits or any explanation of the necessary conditions for achieving these benefits. Furthermore, there seems to be a lack of understanding that time (and financing) is needed in these contexts before these technologies provide benefits.

Recommendations

Based on the findings from the review of the literature and the interviews, we recommend improvements in today's practice of inter-organizational collaboration in e-health.

Our interviews revealed a challenge when the results using maturity metrics were compared. Based on this, the learning from the e-health pilot projects must be embedded in the collaborating organizations' structure using process performance metrics, so that management can effectively control the desired transformation from the as-is process to the future to-be process. Evaluating maturity metrics in the context of existing process structures will help the diagnosis of how process performance can be improved through incremental and innovative technological changes or improvements. This is consistent with the findings from the literature review; inter-organizational management must change from "democratic" in early phases (see the appendix; Concept # 6 "Maturity" ref citations C16, J11, and J21) to stricter in rollout phases (J21) where goals for the improvements are identified.

The CMM [20,26] has been used to describe the development of a systematic process orientation in organizations. Many maturity models can act as instrumental frameworks for management in that they identify practices needed to achieve an organizational goal and act as an assessment tool for tracking the development or maturation within these practices. Thus, maturity models are relevant also here as lenses for the changes that are necessary to initiate and implement to establish the dynamic capabilities needed within e-health. The CMM identifies five levels of maturity: initial (chaotic), repeatable (documented), defined (standardized), managed (with metrics), and optimized (deliberate process management and optimizing). The model predicts that higher levels give better performance. This study suggests that this model or similar maturity models, such as the process and enterprise maturity model [27] or the enterprise architecture maturity [28], can assess the transition or maturation of collaborative networks within public or semi-public e-health innovations through practices that gradually improve process orientation and business model orientation. Venkatesh et al. [28] argued that the enterprise must go through all the stages of maturation, a confirmation of Teece et al.'s argument that dynamic capabilities must be built and that this happens over time.

For long-term performance, the bottom-up approaches posed by individual e-health initiatives have to be met by a structured top-down approach, disciplining the organizations and creating viable business models by imposing an enterprise architecture (EA) to raise the capability maturity from level 2 or 3 (repeatable and managed) as measured in the CMM [20] to level 4 or 5 (managed and optimized).

More research is needed on this area, as migration to EA is a risky project that entails various challenges (for example, two of the Easttowns municipalities had a failed EA project from 2008 to 2011). Service and business modeling must be part of the planning from an early stage [29]. But what level of process orientation can the municipalities absorb? What are the necessary pre-conditions? What is the role of the government here? The role of national politics, laws, and regulations requires further research.

The findings from the interviews show that the national government plays a key role, as illustrated in the model [2]. The national government, through agencies such as the Ministry of Health [Helsedirektoratet], has a significant role in the formation of the national digital information infrastructure. Connected local welfare technology implementations in effect constitute a local digital information infrastructure. To succeed, they have to integrate well with national standards and digital infrastructure (interoperability). The national standards and digital infrastructure should also be designed to allow for new infrastructure to be built on top; see the generativity principle ([30] and the value co-creation principle [31].

Does the national government show enough senior management involvement [9]? Does it provide the necessary frames for the municipalities to fulfill their mission? The main implication of the proposed model in this context is that there is a lack of national governance and coherent policies (see also the appendix, Concept # 4, ref citation C4). This leads to goal divergence and competition between professional groups and between the national government and hospitals and municipalities, and in general, reduced efficiency in the networks.

Further research and model testing on the emergence of process orientation and business or service model orientation in e-health are needed. The Norwegian context could be compared to similar countries following different national strategies.

Limitations

The search design and criteria may have omitted articles that focused on inter-organizational collaborative business process management but used other terms. In screening the articles, we may also have omitted articles that should have been included.

The interviews covered a few selected projects and middle management co-workers in networks that constitute slightly less than 3% of the total number of municipalities in Norway. Interviews and studies in other municipalities, networks, or organizational levels might provide more nuanced findings.

Author statements

Funding

None.

Competing interests

None declared.

Ethical approval

Not required.

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at http://dx.doi.org/10.1016/j.hlpt.2017.02.003

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